
LEC 5: Linear Classifier

Logistic Regression

Logistic regression

- Name is somewhat misleading. Really a technique for classification, not regression.
 - “Regression” comes from fact that we fit a linear model to the feature space.
- Involves a more probabilistic view of classification.

Different ways of expressing probability

- Consider a two-outcome probability space, where:
 - $p(O_1) = p$
 - $p(O_2) = 1 - p = q$
- Can express probability of O_1 as:

	notation	range equivalents		
standard probability	p	0	0.5	1
odds	p / q	0	1	$+\infty$
log odds (logit)	$\log(p / q)$	$-\infty$	0	$+\infty$

Log odds

- Numeric treatment of outcomes O_1 and O_2 is equivalent
 - If neither outcome is favored over the other, then $\log \text{ odds} = 0$.
 - If one outcome is favored with $\log \text{ odds} = x$, then other outcome is disfavored with $\log \text{ odds} = -x$.
- Especially useful in domains where relative probabilities can be miniscule
 - Example: multiple sequence alignment in computational biology

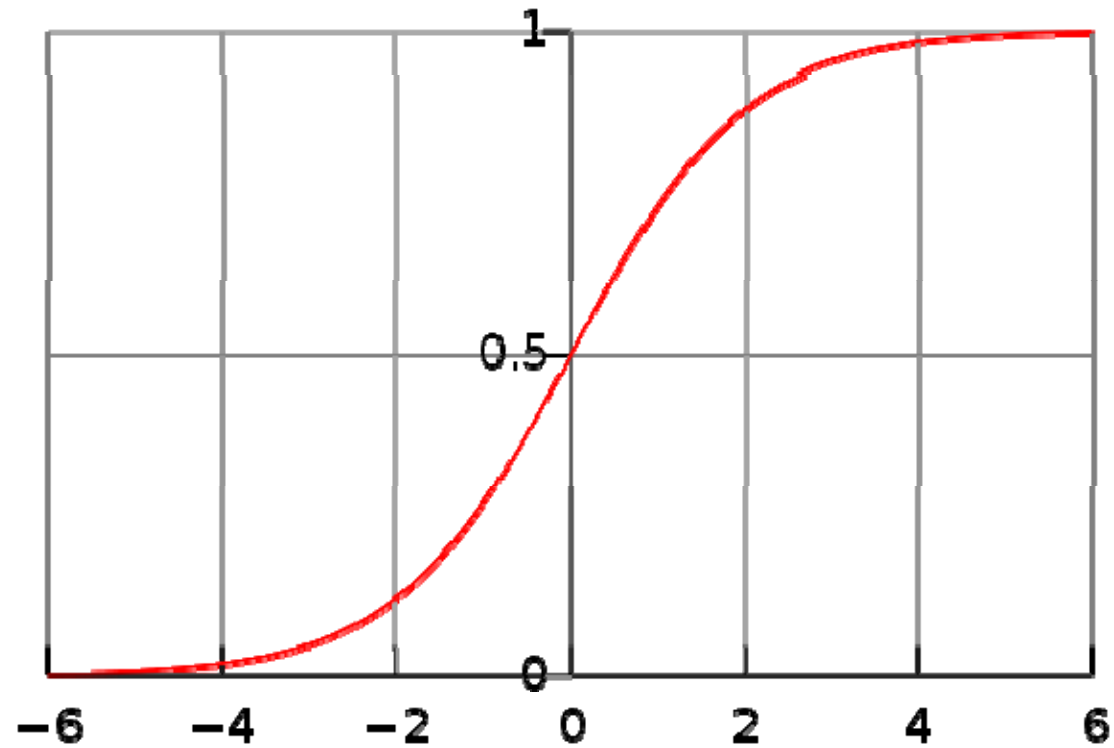
From probability to log odds (and back again)

$$z = \log\left(\frac{p}{1-p}\right) \quad \text{logit function}$$

$$\frac{p}{1-p} = e^z$$

$$p = \frac{e^z}{1+e^z} = \frac{1}{1+e^{-z}} \quad \text{logistic function}$$

Standard logistic function



Logistic regression

- Scenario:
 - A multidimensional feature space (features can be categorical or continuous).
 - Outcome is discrete, not continuous.
 - ◆ We'll focus on case of two classes.
 - It seems plausible that a linear decision boundary (hyperplane) will give good predictive accuracy.

Using a logistic regression model

- Model consists of a vector β in d -dimensional feature space
- For a point $\mathbf{x}$ in feature space, project it onto β to convert it into a real number z in the range $-\infty$ to $+\infty$

$$z = \alpha + \beta \cdot \mathbf{x} = \alpha + \beta_1 x_1 + \cdots + \beta_d x_d$$

- Map z to the range 0 to 1 using the logistic function

$$p = 1/(1 + e^{-z})$$

- Overall, logistic regression maps a point $\mathbf{x}$ in d -dimensional feature space to a value in the range 0 to 1

Using a logistic regression model

- Can interpret prediction from a logistic regression model as:
 - A probability of class membership
 - A class assignment, by applying threshold to probability
 - ◆ threshold represents decision boundary in feature space

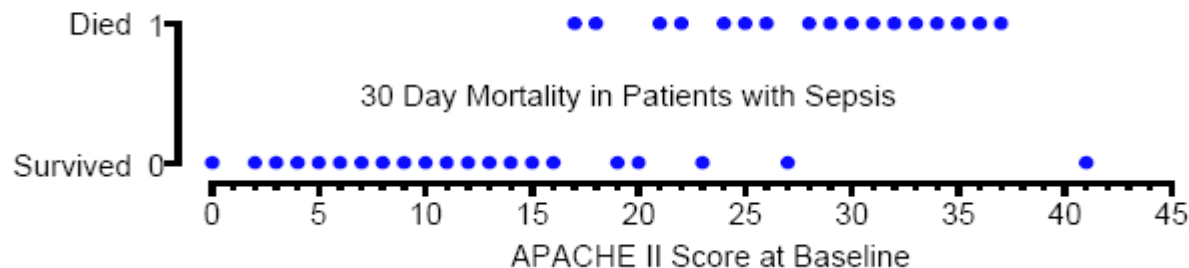
Training a logistic regression model

- Need to optimize β so the model gives the best possible reproduction of training set labels
 - Usually done by numerical approximation of maximum likelihood
 - On really large datasets, may use stochastic gradient descent

Logistic regression in one dimension

a) Example: APACHE II Score and Mortality in Sepsis

The following figure shows 30 day mortality in a sample of septic patients as a function of their baseline APACHE II Score. Patients are coded as 1 or 0 depending on whether they are dead or alive in 30 days, respectively.

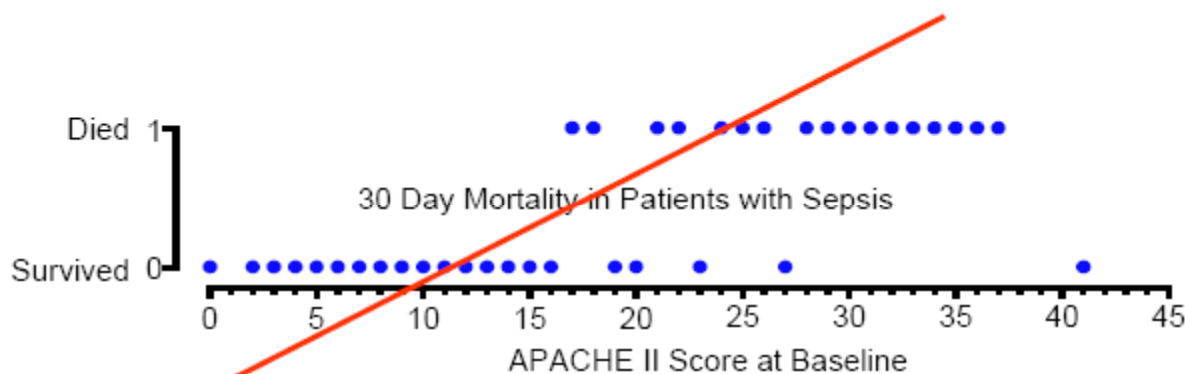


Logistic regression in one dimension

We wish to predict death from baseline APACHE II score in these patients.

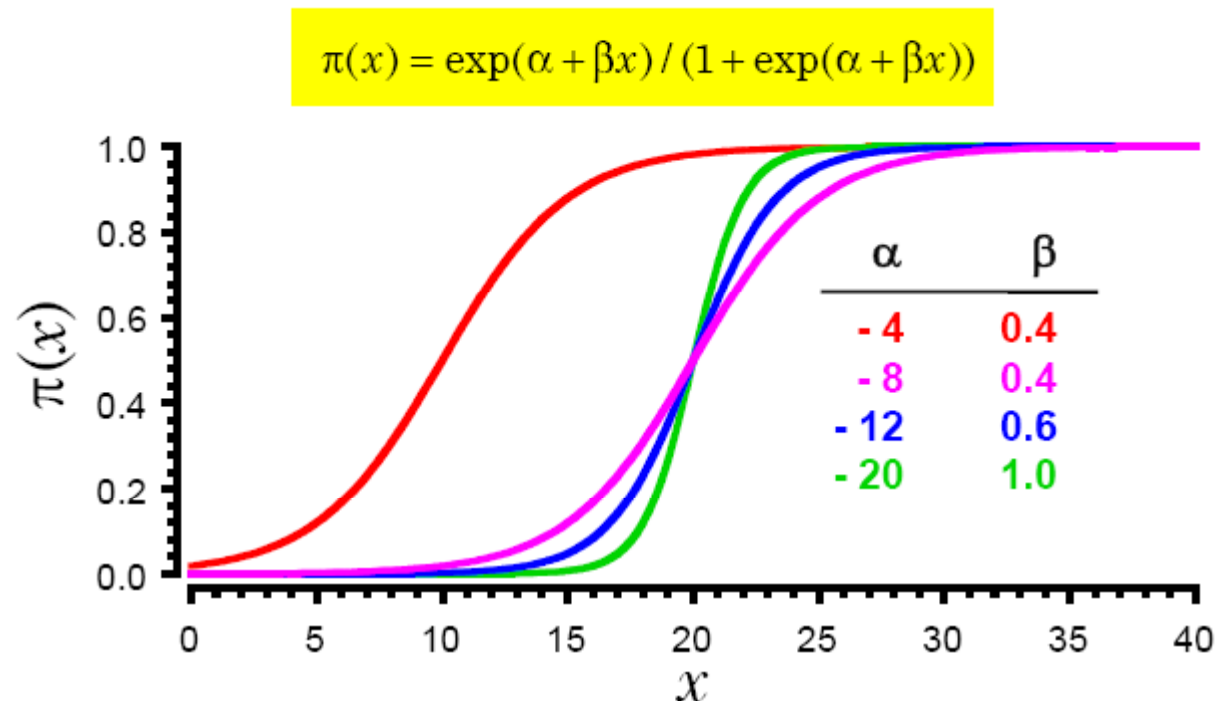
Let $\pi(x)$ be the probability that a patient with score x will die.

Note that linear regression would not work well here since it could produce probabilities less than zero or greater than one.



Logistic regression in one dimension

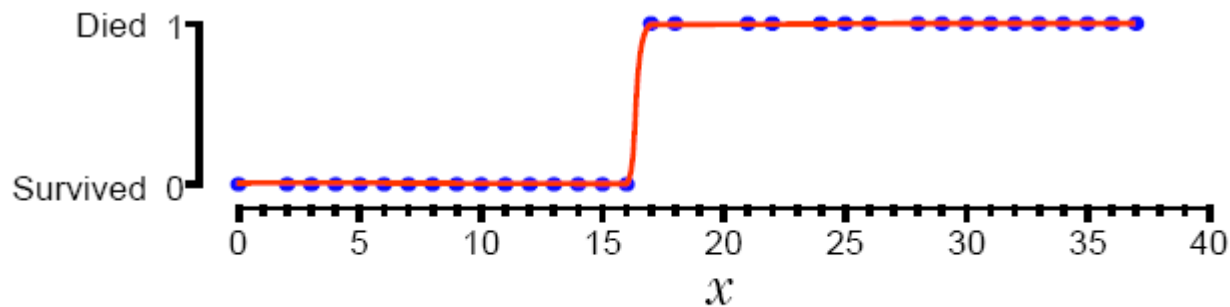
- Parameters control shape and location of sigmoid curve
 - α controls location of midpoint
 - β controls slope of rise



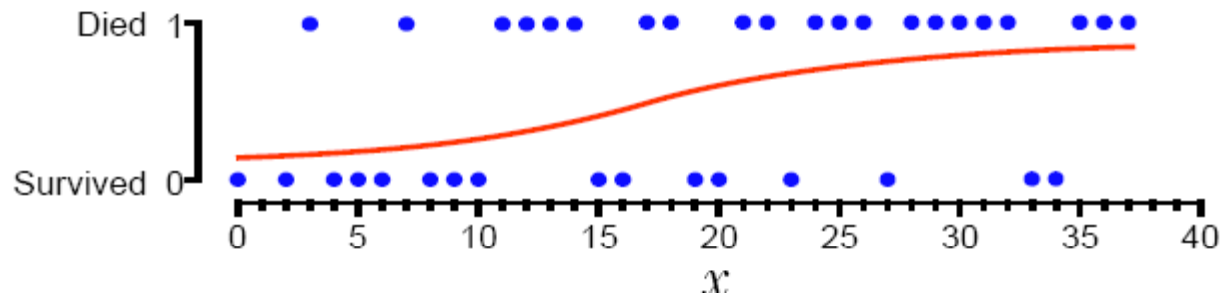
When $x = -\alpha / \beta$, $\alpha + \beta x = 0$ and hence $\pi(x) = 1/(1+1) = 0.5$

Logistic regression in one dimension

Data that has a sharp survival cut off point between patients who live or die should have a large value of β .



Data with a lengthy transition from survival to death should have a low value of β .



IRIS dataset



**Data Set
Characteristics:**

Multivariate

**Number
of
Instances
:**

150

**Features
Characteristics:**

Real

**Number
of
Features:**

4

**Associated
Tasks:**

Classification

**Missing
Values?**

No

Area:

Life

**Date
Donated**

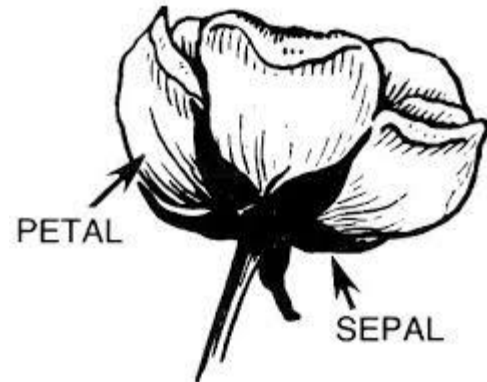
1988-07-01

**Number of
Web Hits:**

1237817

IRIS dataset

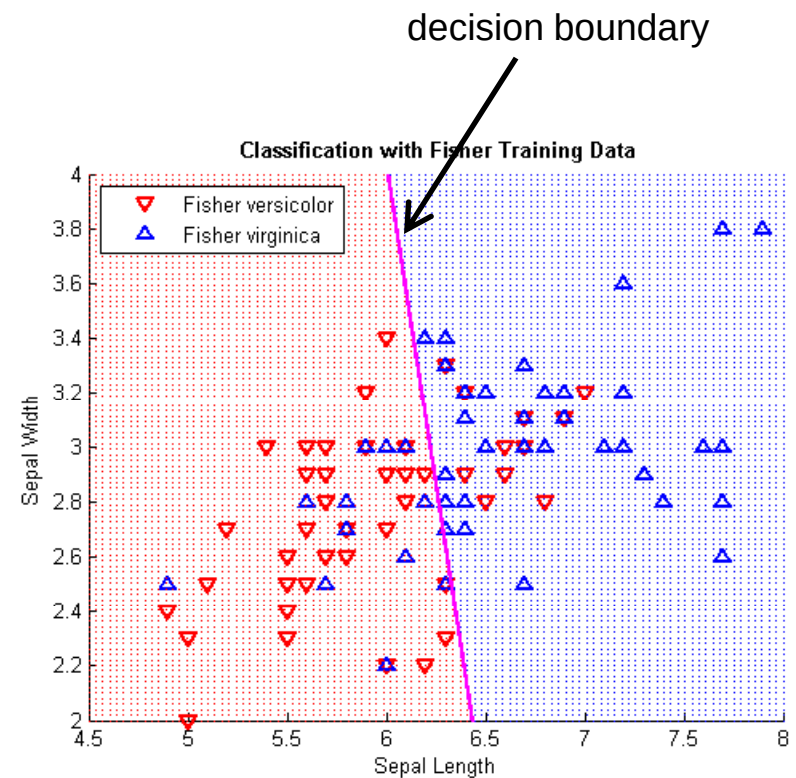
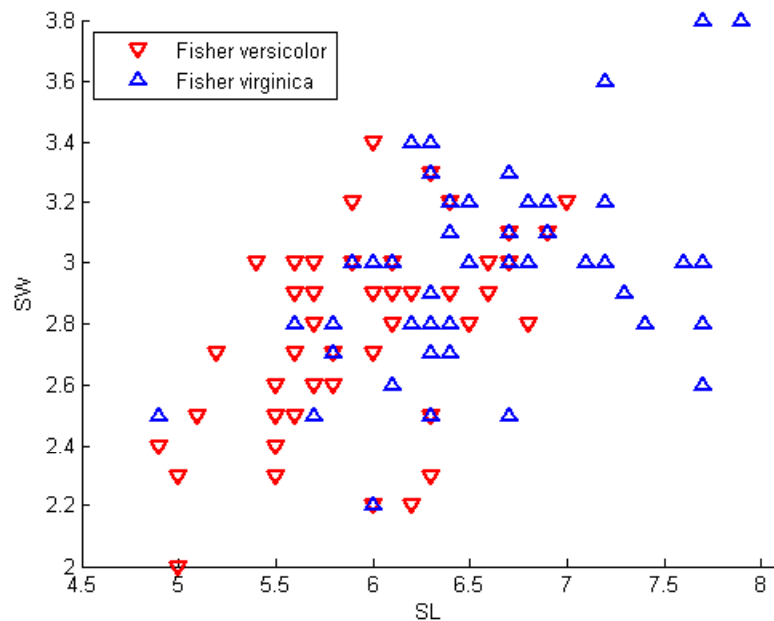
- **Feature Information:**
- 1. sepal length in cm
- 2. sepal width in cm
- 3. petal length in cm
- 4. petal width in cm
- 5. class:
 - Iris Setosa
 - Iris Versicolour
 - Iris Virginica



Logistic regression in two dimensions

Subset of Fisher iris dataset

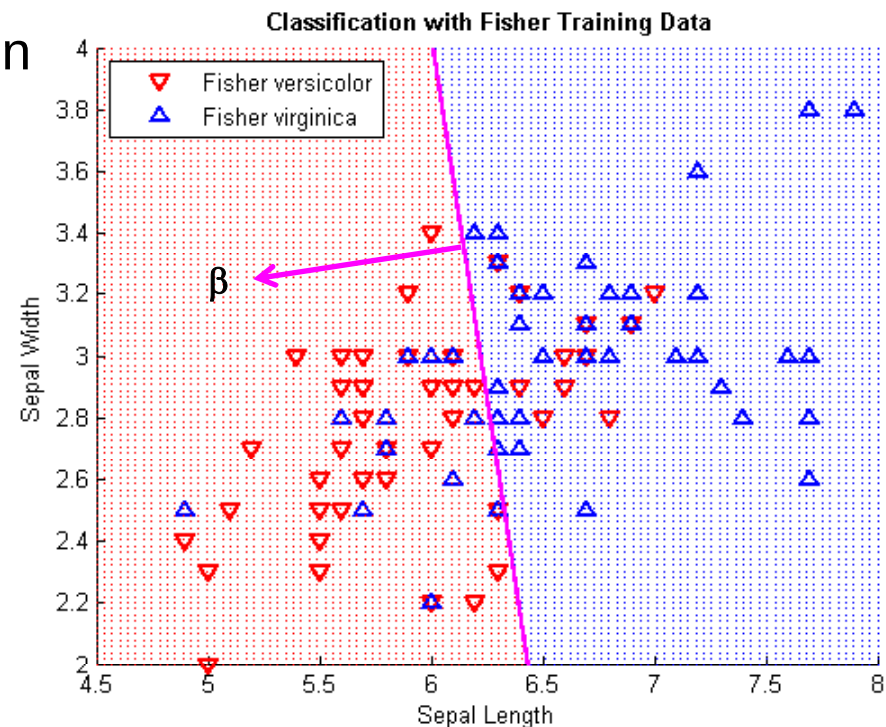
- Two classes
- First two columns (SL, SW)



Logistic regression in two dimensions

Interpreting the model vector of coefficients

- From MATLAB: $B = [13.0460 \quad -1.9024 \quad -0.4047]$
- $\alpha = B(1)$, $\beta = [\beta_1 \ \beta_2] = B(2:3)$
- α , β define location and orientation of decision boundary
 - α is distance of decision boundary from origin
 - decision boundary is perpendicular to β
- magnitude of β defines gradient of probabilities between 0 and 1



Probabilistic interpretation

- Think of $\sigma(f(\mathbf{x}))$ as the **posterior probability** that $y = 1$, i.e. $P(y = 1|\mathbf{x}) = \sigma(f(\mathbf{x}))$
- Hence, if $\sigma(f(\mathbf{x})) > 0.5$ then class $y = 1$ is selected
- Then, after a rearrangement

$$f(\mathbf{x}) = \log \frac{P(y = 1|\mathbf{x})}{1 - P(y = 1|\mathbf{x})} = \log \frac{P(y = 1|\mathbf{x})}{P(y = 0|\mathbf{x})}$$

which is the **log odds ratio**

Maximum Likelihood Estimation

Assume

$$\begin{aligned}p(y = 1|\mathbf{x}; \mathbf{w}) &= \sigma(\mathbf{w}^\top \mathbf{x}) \\p(y = 0|\mathbf{x}; \mathbf{w}) &= 1 - \sigma(\mathbf{w}^\top \mathbf{x})\end{aligned}$$

write this more compactly as

$$p(y|\mathbf{x}; \mathbf{w}) = \left(\sigma(\mathbf{w}^\top \mathbf{x})\right)^y \left(1 - \sigma(\mathbf{w}^\top \mathbf{x})\right)^{(1-y)}$$

Then the likelihood (assuming data independence) is

$$p(\mathbf{y}|\mathbf{x}; \mathbf{w}) \sim \prod_i^N \left(\sigma(\mathbf{w}^\top \mathbf{x}_i)\right)^{y_i} \left(1 - \sigma(\mathbf{w}^\top \mathbf{x}_i)\right)^{(1-y_i)}$$

and the negative log likelihood is

$$L(\mathbf{w}) = - \sum_i^N y_i \log \sigma(\mathbf{w}^\top \mathbf{x}_i) + (1 - y_i) \log(1 - \sigma(\mathbf{w}^\top \mathbf{x}_i))$$

Logistic Regression Loss function

Use notation $y_i \in \{-1, 1\}$. Then

$$P(y = 1|x) = \sigma(f(x)) = \frac{1}{1 + e^{-f(x)}}$$

$$P(y = -1|x) = 1 - \sigma(f(x)) = \frac{1}{1 + e^{+f(x)}}$$

So in both cases

$$P(y_i|x_i) = \frac{1}{1 + e^{-y_i f(x_i)}}$$

Assuming independence, the likelihood is

$$\prod_i^N \frac{1}{1 + e^{-y_i f(x_i)}}$$

and the negative log likelihood is

$$= \sum_i^N \log(1 + e^{-y_i f(x_i)})$$

which defines the **loss function**.

Logistic Regression Learning

Learning is formulated as the optimization problem

$$\min_{\mathbf{w} \in \mathbb{R}^d} \underbrace{\sum_i^N \log \left(1 + e^{-y_i f(\mathbf{x}_i)} \right)}_{\text{loss function}} + \underbrace{\lambda \|\mathbf{w}\|^2}_{\text{regularization}}$$

- For correctly classified points $-y_i f(\mathbf{x}_i)$ is negative, and $\log \left(1 + e^{-y_i f(\mathbf{x}_i)} \right)$ is near zero
- For incorrectly classified points $-y_i f(\mathbf{x}_i)$ is positive, and $\log \left(1 + e^{-y_i f(\mathbf{x}_i)} \right)$ can be large.
- Hence the optimization penalizes parameters which lead to such misclassifications

Logistic regression

- Advantages:
 - Makes no assumptions about distributions of classes in feature space
 - Easily extended to multiple classes (multinomial regression)
 - Natural probabilistic view of class predictions
 - Quick to train
 - Very fast at classifying unknown records
 - Good accuracy for many simple data sets
 - Resistant to overfitting
 - Can interpret model coefficients as indicators of feature importance
- Disadvantages:
 - Linear decision boundary